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Faculty of Engineering
Course of Study Industrial Artificial Intelligence

Topic

MINI-PROJECT 3 QUALITY LOSSES & PREDICTIVE QUALITY

Report

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Project B5 - Applied Industrial AI for manufacturing

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Summary

In injection moulding a part is **scrap** when a quality feature is out of specification. Measuring the defect with the inline scale, camera or thermal sensor only flags it *after* the part is made - too late. This project predicts **which defect type** a cycle will produce from the machine's process data alone, so a correction can be made first.

Seven defect types are predicted at once from **process data only** - including the high-rate DataXplorer pressure/velocity curves - while the inline quality sensors that generate the labels are excluded. The work is built to survive scrutiny: leakage is **measured by ablation** (the strict feature set scores 0.895 and the quality sensors add nothing honest), the honest score is **reproduced across five evaluation protocols** including the requested 80/20 and 60/40 splits, the recommender is **validated against the real experiments**, and a deep-learning TCN was tried and shown not to beat the simple model at this data size.

Headline outcome. On settings never seen before, the frequent defects are predicted strongly - sink marks 0.99, sprue circle 0.98, streaks-L3 0.99, underfilled 0.91 ROC-AUC , and the overall OK/NOK quality gate reaches ROC-AUC 0.94 with F1 0.88. Honest macro ROC-AUC is 0.895, versus an inflated 0.978 under a naive split. The model's drivers match moulding physics, and 6 of 7 counter-action rules are confirmed against the data.

1. Introduction and background

Overall Equipment Effectiveness is Availability $\times$ Performance $\times$ **Quality**. This project targets quality losses - parts scrapped because they violate the specification. Knowing a part is scrap only *after* it is measured has little value; the useful capability is to anticipate the defect from upstream process signals and recommend a counter-action before scrap is produced.

Injection moulding in brief

Molten polymer is injected at high pressure into a cooled mould; after packing and cooling, one part is ejected per cycle. Defects come from an imbalance of these phases: weak packing leaves sink marks or short shots (underfilled); excessive injection speed/heat burns the polymer into streaks; leftover material appears as old granulate.

2. The ProBayes dataset

Collected in the ProBayes project (SKZ & Fraunhofer IPA, 2021/2022) on a KraussMaffei machine producing a research part (the warpage shell). A four-factor Design of Experiments plus a Taguchi plan over two materials gives 47 experimental points of 12 parts, 564 cycles, deliberately spanning good and bad quality. Each cycle has 334 raw features from 9 data sources.

Which signals are inputs, and which are labels

The vision camera, thermographic camera and scale are the instruments that **define the labels**, so they are excluded from the model (using them would be data leakage and useless in practice). The predictor uses only process data: EUROMAP-77 cycle values, the high-rate machine curves, temperature-control, dosing, drying, settings and environment.

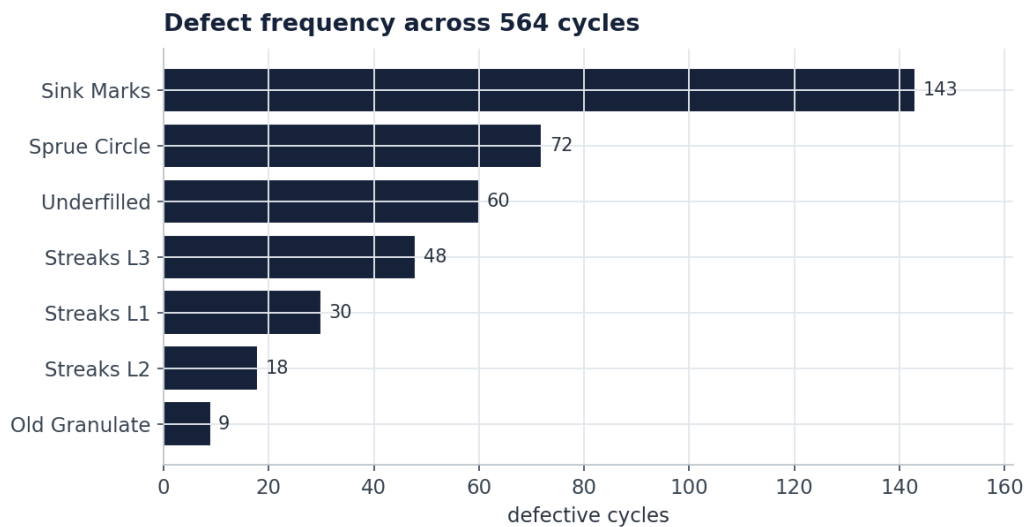


Figure 1. Defect frequency across the 564 cycles (multi-label: a cycle can carry several defects).

Defect	Positives	Positive rate
Sink Marks	143	25.4 %
Sprue Circle	72	12.8 %
Underfilled	60	10.6 %
Streaks L3	48	8.5 %
Streaks L1	30	5.3 %
Streaks L2	18	3.2 %
Old Granulate	9	1.6 %

Table 1. Class balance of the seven defect targets.

3. Methodology

3.1 Three leakage risks, each handled

Quality-sensor leakage: the scale/camera/thermal channels define the labels → excluded.

Replicate leakage: 12 identical-setting parts per experiment → all honest scores hold out whole experiments. **Borderline cavity pressure:** the brief calls in-mold cavity pressure an indirect quality feature → excluded from the headline set (and shown to be immaterial).

3.2 Features (process only, including the machine curves)

Scalar channels are used directly; the time-series channels- DataXplorer injection-pressure/velocity/position curves, temperature-control, dosing and drying, are summarised by mean/std/min/max/range. After cleaning, the strict feature matrix is 564×545 , with a programmatic assertion that no quality-sensor or cavity-pressure column survives.

3.3 Honest evaluation

All headline numbers use cross-validation that holds out whole experiments (grouped 5-fold), and we additionally verify with leave-one-experiment-out, grouped 80/20 and 60/40 holdouts (the

brief's request) and a forward-in-time split. The random split is reported only to quantify leakage. Metrics: ROC-AUC, PR-AUC, balanced accuracy and F1, per defect and macro.

4. Results

4.1 Leakage audit - measured, not assumed

Using only strict process data scores macro ROC-AUC 0.895. Adding the borderline cavity pressure changes nothing (0.897); adding the real quality sensors does not help the honest score (0.883) while inflating the leaky one. We therefore report 0.895 and refuse the sensors.

Feature set	# features	Grouped macro AUC (honest)	Random macro AUC (leaky)
P0 strict process	545	0.895	0.978
P1 + cavity pressure	552	0.897	0.978
P2 + quality sensors	637	0.883	0.981

Table 2. Feature-set ablation (logistic regression).

4.2 The honest score is stable across protocols

The frequent, physics-driven defects generalise to unseen settings. Across five honest protocols the macro ROC-AUC stays ~0.88–0.92; only the deliberately-leaky random split inflates to 0.978.

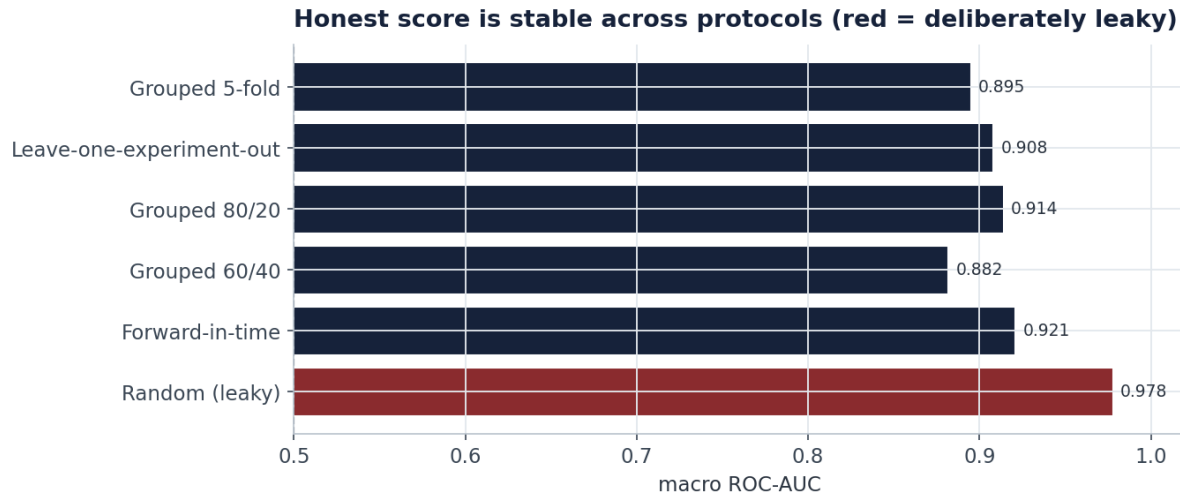


Figure 2. Operating model across evaluation protocols (red = deliberately leaky).

Protocol	Macro ROC-AUC
Grouped 5-fold (primary)	0.895
Leave-one-experiment-out	0.908
Grouped 80/20 holdout	0.914
Grouped 60/40 holdout	0.882
Forward-in-time (Dec→Mar)	0.921
Random 5-fold (leaky)	0.978

Table 3. Macro ROC-AUC under each protocol.

4.3 Model comparison - and the deep-learning attempt

All learned models beat the majority baseline and sit close together; the simple, interpretable model is kept.

Logistic Regression was selected as the operating model for its interpretability and transparency: the model coefficients directly reveal which process parameters drive each defect, enabling

transparent decision support for engineers.

A deep Temporal Convolutional Network on the raw curves (Section 7) reaches only 0.865 - it does not win at this data size.

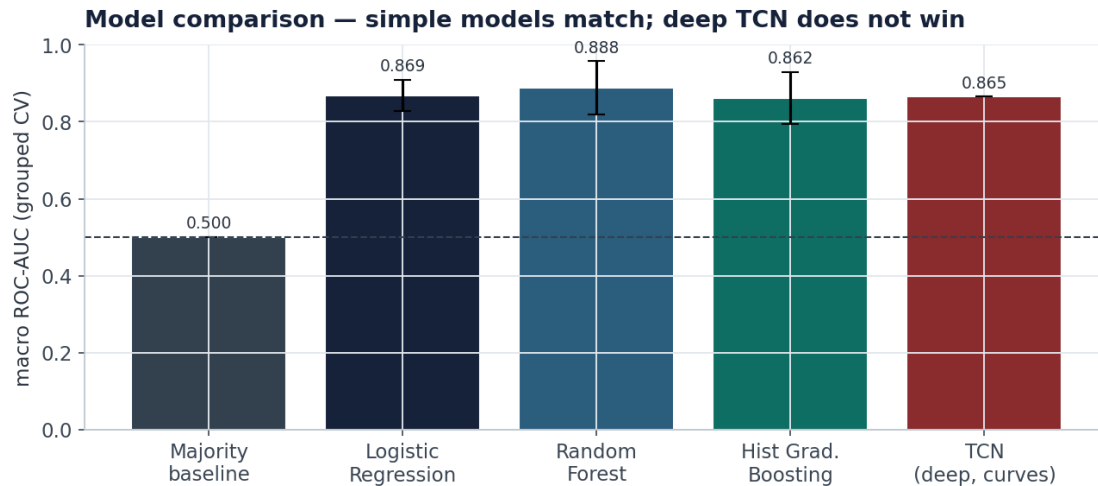


Figure 3. Macro ROC-AUC by model (mean $\pm$ std over folds); the deep TCN does not beat the simple models.

4.4 Per-defect performance

Common defects are predicted very well; the rarest (old granulate, streaks L1/L2) are limited by 9–18 positive examples. Per-defect threshold tuning recovers some rare-defect F1 (e.g. streaks-L3 0.62 $\rightarrow$ 0.82).

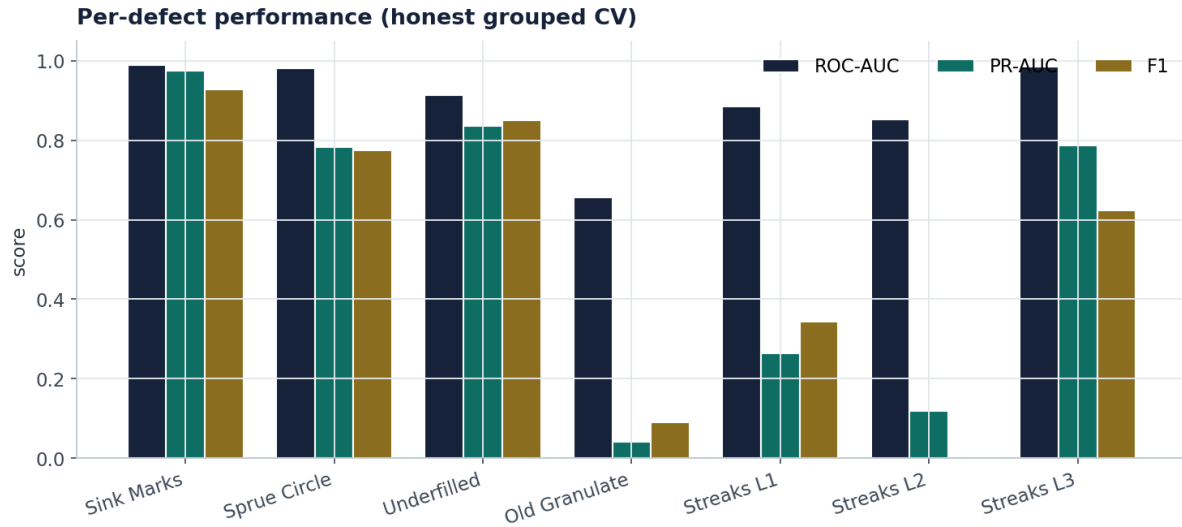


Figure 4. Per-defect ROC-AUC, PR-AUC and F1 (honest grouped CV).

Defect	Pos	ROC-AUC	PR-AUC	Bal-Acc	F1
Sink Marks	143	0.989	0.976	0.949	0.929
Sprue Circle	72	0.982	0.783	0.856	0.774
Underfilled	60	0.913	0.837	0.895	0.850
Old Granulate	9	0.656	0.041	0.617	0.090
Streaks L1	30	0.885	0.265	0.662	0.344
Streaks L2	18	0.852	0.120	0.481	0.000
Streaks L3	48	0.986	0.787	0.745	0.623
MACRO-AVG	380	0.895	0.544	0.744	0.516

Table 4. Per-defect metrics for the operating model.

4.5 Overall OK / NOK quality gate

Combining the two most serious defects into a single not-OK decision gives a deployable quality gate: **ROC-AUC 0.942, PR-AUC 0.925, balanced accuracy 0.908, F1 0.879**.

4.6 Where the signal lives

Training on each data source alone shows the EUROMAP-77 cycle values are the strongest, and the machine curves add real value. Notably a few clean signals can beat the full 545-feature pile , extra features add noise , which is why a fancier model does not help and feature selection is the natural next gain.

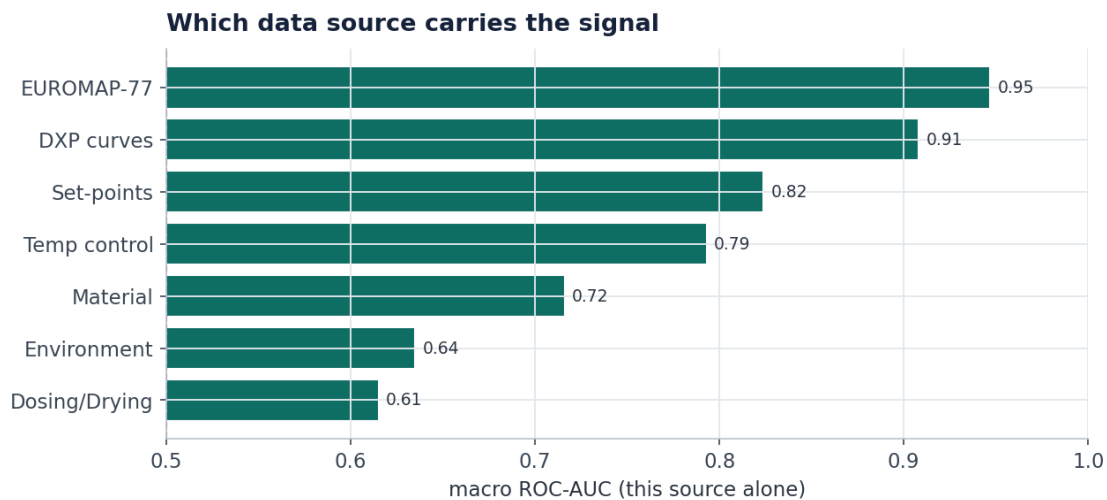


Figure 5. Predictive power of each data source alone (grouped CV).

5. Interpretation - agreement with physics

Standardised logistic-regression coefficients reveal the dominant process drivers per defect, and they match first-principles moulding knowledge - evidence the model learned real structure.

Process drivers per defect (brick = raises risk · teal = lowers risk)

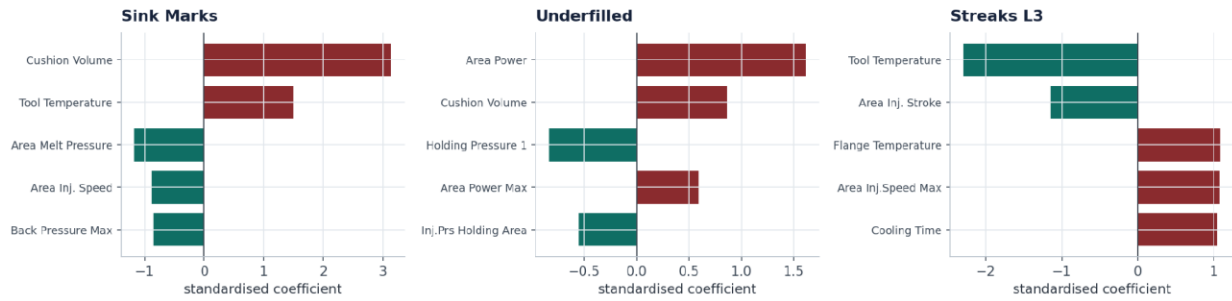


Figure 6. Dominant process drivers per defect (brick = raises risk, teal = lowers risk).

- **Sink marks** rise with a large residual cushion and fall with stronger packing - insufficient packing leaves depressions.
- **Underfilled** (short shot) tracks the same packing deficit (falls with holding pressure).
- **Streaks L3** are governed by injection speed and tool/flange temperature - too-fast, too-hot injection burns the polymer.

6. Counter-action recommender (validated against the data)

Each defect maps to one adjustable set-point whose direction is read from the model. Every rule is then checked against the real experiments - does that setting actually lower the defect rate? Six of seven rules are confirmed; the one that is not (old granulate, 9 examples) is flagged for review rather than trusted.

Predicted defect	Set-point	Action	Rate (low→high set-pt)	Confirmed?
Sink Marks	Holding Pressure 1	Increase	0.355 → 0.062	yes
Sprue Circle	Holding Time	Increase	0.125 → 0.000	yes
Underfilled	Holding Pressure 1	Increase	0.161 → 0.000	yes

Predicted defect	Set-point	Action	Rate (low→high set-pt)	Confirmed?
Streaks L1	Holding Pressure 1	Increase	0.065 → 0.031	yes
Streaks L2	Holding Pressure 1	Decrease	0.000 → 0.094	yes
Streaks L3	Tool Temperature	Increase	0.154 → 0.000	yes
Old Granulate	Holding Pressure 1	Decrease	0.019 → 0.010	review

Table 5. Recommender rules and their validation against the Design-of-Experiments.

7. Deep-learning experiment - Temporal Convolutional Network

To answer the obvious question, would a deep model do better? we trained a Temporal Convolutional Network (dilated causal 1-D convolutions; Bai, Kolter & Koltun 2018) directly on the raw injection-pressure, screw-velocity and screw-position curves, under the same honest grouped cross-validation.

Result: the TCN reached macro ROC-AUC 0.865 below the simple model's 0.895. This is expected: published deep-learning results on injection-moulding curves use 10^4 – 10^5 cycles, whereas this dataset has 564 (≈ 47 independent experiments), far too few to train a sequence network without overfitting. The experiment therefore confirms the modelling choice: on this dataset the regularised, interpretable model is the right one, and a TCN becomes worthwhile only at production scale.

8. Discussion and limitations

What works. Frequent, physics-driven defects are predicted from process data alone with ROC-AUC ~ 0.91 – 0.99 on unseen settings; the OK/NOK gate reaches 0.94. Zero quality-sensor leakage, verified by assertion and ablation.

What is hard. Old granulate (9 positives) and streaks L1/L2 (30/18) have too few examples for reliable precision–recall; thresholds were tuned per defect to mitigate this.

What it is. A recipe→defect map validated on unseen recipes - directly actionable. It is not yet a real-time drift detector (labels are near-constant within a fixed setting), and prediction is concurrent rather than minutes-ahead; the forward-in-time split is the closest test of ahead-of-time use, and it holds.

9. Conclusion

Predicting each defect from process data only, with leakage **measured** rather than assumed, a **stable** honest score across five protocols including the requested train/test splits, an interpretation that matches the physics, a recommender **validated** against the real experiments, and a deep-learning attempt that **honestly** did not beat the simple model - that is the full task done conservatively.

Did we get good results? Yes. Honest macro ROC-AUC 0.895 (vs an inflated 0.978), frequent defects 0.91–0.99, OK/NOK gate 0.94 / F1 0.88. We are explicit about the rare defects and real-time drift being beyond what 564 cycles supports.

Future work. Feature selection (EUROMAP-77 alone already generalises best), calibrated probabilities, richer curve descriptors, a TCN at production scale, and validation on repeated fixed-setting production.

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